LESSON VI

SQL INTRODUCTION

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LEARNING OBJECTIVES

1. Define SQL
2. Enumerate the capabilities of SQL
3. Identify the different SQL commands and execute query using SQL Select statement

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SQL INTRODUCTION

- SQL is a standard language for accessing and manipulating databases.

What is SQL?

- SQL stands for Structured Query Language
- SQL lets you access and manipulate databases
- SQL became a standard of the ANSI in 1986, and of the International Organization for Standardization (ISO) in 1987
- SQL is a special purpose programming language designed for managing data in relational database management systems (RDBMS).

What Can SQL do?

- SQL can execute queries against a database.
- SQL can retrieve data from a database
- SQL can insert records in a database
- SQL can update records in a database
- SQL can delete records from a database
- SQL can create new databases
- SQL can create new tables in a database
- SQL can create stored procedures in a database
- SQL can create views in a database
- SQL can set permissions on tables, procedures, and views

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Objectives of SQL

Ideally, a database language should allow a user to:

- create the database and relation structures;
- perform basic data management tasks, such as the insertion, modification, and deletion of data from the relations;
- perform both simple and complex queries.

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Two major components:

Data Definition Language (DDL) for defining the database structure and controlling access to the data;

- **CREATE** creates an object (a table, for example) in the database.
- **DROP** deletes an object in the database, usually irretrievably.
- **ALTER** modifies the structure an existing object in various ways—for example, adding a column to an existing table.

Data Manipulation Language (DML) for retrieving and updating data.

Operation	SQL	Description
Create	INSERT INTO	inserts new data into a database
Read (Retrieve)	SELECT	extracts data from a database
Update	UPDATE	updates data in a database
Delete (Destroy)	DELETE	deletes data from a database

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Using SQL in Your Web Site

To build a web site that shows data from a database, you will need:

- An RDBMS database program (i.e. MS Access, SQL Server, MySQL)
- To use a server-side scripting language, like PHP or ASP
- To use SQL to get the data you want
- To use HTML / CSS to style the page



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RDBMS

- RDBMS stands for Relational Database Management System.
- RDBMS is the basis for SQL, and for all modern database systems such as MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access.
- The data in RDBMS is stored in database objects called tables. A table is a collection of related data entries and it consists of columns and rows.

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RDBMS

- **Fields -** A field is a column in a table that is designed to maintain specific information about every record in the table.

  - is broken up into smaller entities.

- **Record** – also called a row, is each individual entry that exists in a table.
- **Column** - is a vertical entity in a table that contains all information associated with a specific field in a table.

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SQL Syntax

- Database Tables

A database most often contains one or more tables. Each table is identified by a name (e.g. “tblCustomers" or “tblOrders"). Tables contain records (rows) with data.

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

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SQL Statements

The following SQL statement **selects** all the records in the “tblCustomers” table:

Example

```
SELECT * FROM tblCustomers;
```

- **Semicolon** is the standard way to separate each SQL statement in database systems that allow more than one SQL statement to be executed in the same call to the server.



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Some of The Most Important SQL Commands

- **SELECT** - extracts data from a database
- **UPDATE** - updates data in a database
- **DELETE** - deletes data from a database
- **INSERT INTO** - inserts new data into a database
- **CREATE DATABASE** - creates a new database

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Some of The Most Important SQL Commands

- ALTER DATABASE - modifies a database
- CREATE TABLE - creates a new table
- ALTER TABLE - modifies a table
- DROP TABLE - deletes a table
- CREATE INDEX - creates an index (search key)
- DROP INDEX - deletes an index

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SQL SELECT Statement

- The SELECT statement is used to select data from a database.
- The data returned is stored in a result table, called the result-set.
- **SELECT Syntax**

```
SELECT column1, column2, ...
FROM table_name;
```

or

```
SELECT * FROM table_name;
```

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Demo Database

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

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SELECT Column Example

The following SQL statement selects the "CustomerName" and "City" columns from the “tblCustomers" table:
Example:

```
SELECT CustomerName, City FROM tblCustomers;
```

SELECT * Example
The following SQL statement selects all the columns from the “tblCustomers" table:
Example

```
SELECT * FROM tblCustomers;
```

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tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
2	Jackson	John	London	036163625

Example #1: SELECT * FROM tblCUSTOMER;

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
2	Jackson	John	London	036163625

Example #2: SELECT CustomerId, LastName, FirstName FROM tblCUSTOMER;

CustomerID	Lastname	Firstname
1	Smith	John
2	Jackson	John

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SQL SELECT DISTINCT Statement

- The SELECT DISTINCT statement is used to return only distinct (different) values.

SELECT DISTINCT Syntax

```
SELECT DISTINCT column1, column2
FROM table_name;
```



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tblCUSTOMER

CustomerID	Lastname	Firstname
1	Smith	John
2	Anderson	Michael
3	Jackson	John

Example: SELECT DISTINCT Firstname FROM tblCUSTOMER;

Firstname
John
Michael



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SQL WHERE Clause

- The WHERE clause is used to filter records.
- The WHERE clause is used to extract only those records that fulfill a specified condition.

WHERE Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```



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tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
2	Jackson	John	London	036163625

Example #1 (Numeric Fields):

```
SELECT Lastname,Firstname,ContactNo FROM tblCUSTOMER
WHERE CustomerID = 1;
```

Lastname	Firstname	ContactNo
Smith	John	028275787



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tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
2	Jackson	John	London	036163625

Example #2 (Text Fields):

```
SELECT * FROM tblCUSTOMER WHERE Lastname = 'JACKSON';
```

CustomerID	Lastname	Firstname	Address	ContactNo
2	Jackson	John	London	036163625



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Operators in The WHERE Clause

Operator	Description
=	Equal
<>	Not equal. Note: In some versions of SQL this operator may be written as !=
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
BETWEEN	Between an inclusive range (ie. BETWEEN 1 AND 3)
LIKE	Search for a pattern (ie. LIKE '%jo%')
IN	To specify multiple possible values for a column. ie. IN('California','New York')

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SQL AND, OR and NOT Operators

- The WHERE clause can be combined with AND, OR, and NOT operators.

- The AND and OR operators are used to filter records based on more than one condition:
  - The AND operator displays a record if all the conditions separated by AND are TRUE
  - The OR operator displays a record if any of the conditions separated by OR is TRUE.

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- The NOT operator displays a record if the condition(s) is NOT TRUE.

tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
2	Jackson	John	London	036163625
3	Anderson	Michael	California	019274161

Example #1 (AND OPERATOR): SELECT * FROM tblCUSTOMER WHERE Lastname = 'Smith' AND Address = 'California';

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787



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tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
2	Jackson	John	London	036163625
3	Anderson	Michael	California	019274161

Example #2 (OR OPERATOR): SELECT * FROM tblCUSTOMER WHERE Lastname = 'Smith' OR Address = 'California';

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	John	California	028275787
3	Anderson	Michael	California	019274161

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SQL AND, OR and NOT Syntax

AND Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE condition1 AND
condition2 AND condition3 ...;
```

OR Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE condition1 OR condition2
OR condition3 ...;
```

NOT Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE NOT condition;
```

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AND Example

Example

```
SELECT * FROM Customers
WHERE Country='Germany' AND
City='Berlin';
```

OR Example

Example

```
SELECT * FROM Customers
WHERE City='Berlin' OR
City='München';
```

Example

```
SELECT * FROM Customers
WHERE Country='Germany'
OR Country='Spain';
```



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NOT Example

Example

```
SELECT * FROM Customers WHERE NOT
Country='Germany';
```

Combining AND, OR and NOT

Example

```
SELECT * FROM Customers
 WHERE Country='Germany' AND (City='Berlin' OR
City='München');
```

Example

```
SELECT * FROM Customers
WHERE NOT Country='Germany' AND NOT
Country='USA';
```

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SQL ORDER BY Keyword

- The ORDER BY keyword is used to sort the result-set in ascending or descending order.
- The ORDER BY keyword sorts the records in ascending order by default. To sort the records in descending order, use the DESC keyword.

ORDER BY Syntax

```
SELECT column1, column2, ...
FROM table_name
ORDER BY column1, column2, ... ASC|DESC;
```

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tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	Anne	California	028275787
2	Jackson	Mary	London	036163625
3	Anderson	Michael	California	019274161
4	Fernandez	Jenny	New York	028173719

Example #1: SELECT * FROM tblCUSTOMER ORDER BY Address, Lastname;

CustomerID	Lastname	Firstname	Address	ContactNo
3	Anderson	Michael	California	019274161
1	Smith	Anne	California	028275787
2	Jackson	Mary	London	036163625
4	Fernandez	Jenny	New York	028173719

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tblCUSTOMER

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	Anne	California	028275787
2	Jackson	Mary	London	036163625
3	Anderson	Michael	California	019274161
4	Fernandez	Jenny	New York	028173719

Example #2: SELECT * FROM tblCUSTOMER ORDER BY Address, Lastname DESC;

CustomerID	Lastname	Firstname	Address	ContactNo
1	Smith	Anne	California	028275787
3	Anderson	Michael	California	019274161
2	Jackson	Mary	London	036163625
4	Fernandez	Jenny	New York	028173719

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SQL LIKE Operator

- The LIKE operator is used in a WHERE clause to search for a specif ied pattern in a column. The LIKE operator is used to search for a specified pattern in a column.

Syntax

```
SELECT column_name(s) FROM table_name WHERE
column_name LIKE pattern;
```



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SQL LIKE Operator

tblSTUDENT

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Manalo	Erwin	Imus	091386168

Example #1: The following SQL statement selects all students with Lastname ending with the pattern 'LO':

SELECT * FROM tblSTUDENT WHERE LASTNAME LIKE '%LO'

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
3	Manalo	Erwin	Imus	091386168



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SQL LIKE Operator

Example #2: The following SQL statement selects all students with Lastname starting with the pattern 'P':

SELECT * FROM tblSTUDENT WHERE LASTNAME LIKE 'P%'

StudentID	Lastname	Firstname	Address	ContactNo
2	Panganiban	Mary	Naic	092817262

Example #3: The following SQL statement selects all students with Lastname containing with the pattern 'B':

SELECT * FROM tblSTUDENT WHERE LASTNAME LIKE '%B%'

StudentID	Lastname	Firstname	Address	ContactNo
2	Panganiban	Mary	Naic	092817262

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SQL Wildcards

A wildcard character can be used to substitute for any other character(s) in a string. In SQL, wildcard characters are used with the SQL LIKE operator.

Wildcard	Description
%	A substitute for zero or more characters
_	A substitute for a single character
[*charlist*]	Sets and ranges of characters to match
[^*charlist*] or [!*charlist*]	Matches only a character NOT specified within the brackets

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SQL Wildcards

tblSTUDENT

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Manalo	Erwin	Imus	091386168

Example #1 (Using _ Wildcard): The following SQL statement selects all students with Lastname starting with any character, followed by "anganiban":

SELECT * FROM tblSTUDENT WHERE Lastname LIKE '_anganiban'

StudentID	Lastname	Firstname	Address	ContactNo
2	Panganiban	Mary	Naic	092817262

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SQL Wildcards

Example #2 (Using [charlist] Wildcard): The following SQL statement selects all students with Firstname starting with "c" or "m":

SELECT * FROM tblSTUDENT WHERE Firstname LIKE '[cm]%'

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262

Example #3 (Using [^charlist] Wildcard: The following SQL statement selects all students with Firstname NOT starting with "c" or "m":

SELECT * FROM tblSTUDENT WHERE Firstname LIKE '[^cm]%'

StudentID	Lastname	Firstname	Address	ContactNo
3	Manalo	Erwin	Imus	091386168

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SQL IN Operator

The IN operator allows you to specify multiple values in a WHERE clause.
Syntax:

SELECT column_name(s) FROM table_name WHERE column_name IN (value1,value2,...)

tblSTUDENT

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Manalo	Erwin	Imus	091386168

Example: SELECT * FROM tblSTUDENT WHERE Address IN ('Bacoor','Imus')

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
3	Manalo	Erwin	Imus	091386168

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SQL BETWEEN Operator

The BETWEEN operator selects values within a range. The values can be numbers, text, or dates.

Syntax:

```
SELECT column_name(s) FROM table_name WHERE column_name
BETWEEN value1 AND value2;
```



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SQL BETWEEN Operator

tblSTUDENT

StudentID	Lastname	Firstname	Address	Birthdate
1	Manalo	Chrystel	Bacoor	1990-10-22
2	Panganiban	Mary	Naic	1999-05-10
3	Manalo	Erwin	Imus	1993-01-20

Example #1: SELECT * FROM tblSTUDENT WHERE StudentID BETWEEN 2 AND 3

StudentID	Lastname	Firstname	Address	Birthdate
2	Panganiban	Mary	Naic	1999-05-10
3	Manalo	Erwin	Imus	1993-01-20

Example #2: SELECT * FROM tblSTUDENT WHERE Birthdate BETWEEN '10-22-1990'AND '01-20-1993'

StudentID	Lastname	Firstname	Address	Birthdate
1	Manalo	Chrystel	Bacoor	1990-10-22
3	Manalo	Erwin	Imus	1993-01-20

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The SQL INSERT INTO Statement

- The INSERT INTO statement is used to insert new records in a table.

INSERT INTO Syntax

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

```
INSERT INTO table_name
VALUES (value1, value2, value3, ...);
```

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INSERT INTO Example

The following SQL statement inserts a new record in the "Customers" table:

Example

```
INSERT INTO Customers (CustomerName, ContactName,
Address, City, PostalCode, Country) VALUES ('Cardinal',
'Tom B.  Erichsen',  'Skagen 21', 'Stavanger', '4006', 'Norway');
```

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Demo Database

Below is a selection from the "Customers" table in the Northwind sample database:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
89	White Clover Markets	Karl Jablonski	305 - 14th Ave. S. Suite 3B	Seattle	98128	USA
90	Wilman Kala	Matti Karttunen	Keskuskatu 45	Helsinki	21240	Finland
91	Wolski	Zbyszek	ul. Filtrowa 68	Walla	01-012	Poland
92	Cardinal	Tom B. Erichsen	Skagen 21	Stavanger	4006	Norway

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Insert Data Only in Specified Columns

The following SQL statement will insert a new record, but only insert data in the "CustomerName", "City", and "Country" columns (CustomerID will be updated automatically):

Example

```
INSERT INTO Customers (CustomerName, City, Country) VALUES
('Cardinal', 'Stavanger', 'Norway');
```

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SQL NULL Values

What is a NULL Value?

A field with a NULL value is a field with no value.

If a field in a table is optional, it is possible to insert a new record or update a record without adding a value to this field. Then, the field will be saved with a NULL value.

Note: A NULL value is different from a zero value or a field that contains spaces. A field with a NULL value is one that has been left blank during record creation!

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Demo Database

The selection from the "Customers" table will now look like this:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
89	White Clover Markets	Karl Jablonski	305 - 14th Ave. S. Suite 3B	Seattle	98128	USA
90	Wilman Kala	Matti Karttunen	Keskuskatu 45	Helsinki	21240	Finland
91	Wolski	Zbyszek	ul. Filtrowa 68	Walla	01-012	Poland
92	Cardinal	null	null	Stavanger	null	Norway

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How to Test for NULL Values?

It is not possible to test for NULL values with comparison operators, such as =, <, or <>.

We will have to use the IS NULL and IS NOT NULL operators instead.

IS NULL Syntax

```
SELECT column_names
FROM table_name
WHERE column_name IS NULL;
```

IS NOT NULL Syntax

```
SELECT column_names
FROM table_name
WHERE column_name IS NOT NULL;
```



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INSERT INTO Example

tblSTUDENT

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Atienza	John Mike	Trece	091282616
4	Perido	Jennifer	Indang	090725161

Example #1: INSERT INTO tblSTUDENT VALUES ('Manalo','Ron','Imus','091525261')

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Atienza	John Mike	Trece	091282616
4	Perido	Jennifer	Indang	090725161
5	Manalo	Ron	Imus	091525261

Example #2: INSERT INTO tblSTUDENT (Lastname,Firstname,Address) VALUES ('Carandang','Khimberly','Rosario')

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Atienza	John Mike	Trece	091282616
4	Perido	Jennifer	Indang	090725161
5	Manalo	Ron	Imus	091525261
6	Carandang	Khimberly	Rosario	NULL

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SQL UPDATE Statement

The UPDATE statement is used to modify the existing records in a table.

UPDATE Syntax

```
UPDATE table_name
SET column1 = value1, column2 = value2, ... WHERE condition;
```

Note: Be careful when updating records in a table! Notice the WHERE clause in the UPDATE statement. The WHERE clause specifies which record(s) that should be updated. If you omit the WHERE clause, all records in the table will be updated!

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Customers

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
101	Alfreds Futterkiste	alfred Schmidt	Obere Str. 57	Frank	12209	Germany
201	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
301	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
401	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
501	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

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UPDATE Table

The following SQL statement updates the f irst customer (CustomerID = 101) with a new contact person and a new city.

Example

```
UPDATE Customers
SET ContactName = 'Alfred Schmidt', City= 'Frankfurt'
WHERE CustomerID = 101;
```

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OUTPUT

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
101	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
201	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
301	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
401	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
501	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

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UPDATE Multiple Records

It is the WHERE clause that determines how many records will be updated.

The following SQL statement will update the contactname to "Juan" for all records where country is "Mexico":

Example

```
UPDATE Customers
SET ContactName='Juan'      WHERE
Country='Mexico';
```



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OUTPUT

The selection from the "Customers" table will now look like this:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
2	Ana Trujillo Emparedados y helados	Juan	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Juan	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

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UPDATE Warning !

Be careful when updating records. If you omit the WHERE clause, ALL records will be updated!

Example

```
UPDATE Customers
SET ContactName='Juan';
```



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OUTPUT

The selection from the "Customers" table will now look like this:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Juan	Obere Str. 57	Frankfurt	12209	Germany
2	Ana Trujillo Emparedados y helados	Juan	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Juan	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Juan	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Juan	Berguvsvägen 8	Luleå	S-958 22	Sweden

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Update Statement

tblSTUDENT

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Atienza	John Mike	Trece	091282616
4	Perido	Jennifer	Indang	090725161

Example: UPDATE tblSTUDENT SET Address = 'Imus' ,ContactNo = '091292727' WHERE StudentID = 3

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Atienza	John Mike	Imus	091292727
4	Perido	Jennifer	Indang	090725161

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SQL DELETE Statement

- The DELETE statement is used to delete existing records in a table.

DELETE Syntax

```
DELETE FROM table_name WHERE condition;
```

Note: Be careful when deleting records in a table! Notice the WHERE clause in the DELETE statement. The WHERE clause specifies which record(s) should be deleted. If you omit the WHERE clause, all records in the table will be deleted!

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Demo Database Delete

tblSTUDENT

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262
3	Atienza	John Mike	Trece	091282616
4	Perido	Jennifer	Indang	090725161

Example: DELETE FROM tblSTUDENT WHERE StudentID > 2

StudentID	Lastname	Firstname	Address	ContactNo
1	Manalo	Chrystel	Bacoor	091028172
2	Panganiban	Mary	Naic	092817262

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SQL DELETE Example

Example

```
DELETE FROM Customers WHERE CustomerName='Alfreds Futterkiste';
```

Output:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

Edit with WPS Office

Delete All Records

It is possible to delete all rows in a table without deleting the table. This means that the table structure, attributes, and indexes will be intact:

Syntax

```
DELETE FROM table_name;
```

Example

```
DELETE FROM Customers;
```



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